# Explaining Crop-Yield Anomalies with Multi-Method Interpretable Machine Learning

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Abstract—Yield prediction alone does not explain why particular crop–state–year outcomes fall below their long-run expectation. This study combines detrended low-yield anomaly screening with a multi-method interpretable machine-learning protocol to identify which extreme-weather driver groups are consistently emphasized by a fitted residual model. We use a U.S. crop–state–year panel for Barley, Canola, Oats, and Wheat from 1990–2025, with USDA yield records and NASA POWER weather features. Linear detrending within crop–state series flags 214 of 1,257 observations as low-yield anomalies using $z < -1.0$. ExtraTrees yield models retain strong forward-time skill, while the residual model is treated as a diagnostic surface rather than a high-skill anomaly-magnitude forecaster. We compare SHAP, grouped SHAP, group permutation, group ablation, ALE curves, and selected LIME case studies. Heat is the strongest global model-reliance signal, while anomaly-focused explanations show co-leading heat and drought signals across methods. The contribution is a robust grouped-XAI diagnostic protocol for cautious model-based interpretation rather than causal event attribution.

Index Terms—crop yield, yield anomaly, explainable machine learning, SHAP, permutation importance, drought, heat stress

## I. Introduction

Crop-yield machine learning is often evaluated by average predictive accuracy, but decision-makers also need to understand low-yield anomaly years. A model that predicts yield levels well can still leave open a diagnostic question: which physical weather drivers does the fitted model associate with below-trend crop outcomes? This distinction matters because prediction and explanation can diverge, especially when weather variables are correlated and yield anomalies are rare.

This paper reframes the task as multi-method model explanation. We first detrend yield within crop–state series and flag unusually low residual years. We then train a weather-yield residual model and compare explanation methods that operate at feature, driver-group, response-curve, and selected event levels. The central question is: which extreme-weather driver groups are consistently emphasized by the fitted residual model, and are those conclusions stable across several interpretable machine-learning methods?

The study makes three contributions. First, it combines crop–state yield detrending with low-yield anomaly screening to separate long-run production trends from short-run weather-sensitive deviations. Second, it implements a shared driver-group XAI protocol that compares feature-level SHAP, grouped SHAP, group permutation, group ablation, ALE response curves, and selected LIME explanations. Third, it evaluates method agreement at global and anomaly-focused levels, allowing driver claims to be stated more cautiously when several methods identify the same weather-driver group.

The analysis avoids causal overclaiming. Terms such as “driver”, “attribution”, and “explanation” are used in a model-based diagnostic sense. Agreement across methods means that the fitted model repeatedly identifies the same driver group; it does not prove a real-world causal mechanism or formal climate-event attribution. This framing is useful for decision support because it distinguishes between forecasting yield levels and diagnosing the fitted model’s behavior on below-trend events.

## II. Related Work

Machine-learning yield studies show that weather and climate variables can support crop prediction across regions and crops [1], [2], [3], [4]. A related literature links yield anomalies to weather extremes, including drought, heat, excess water, and cold stress [5], [6], [7], [8], [9], [10]. Because technological and management trends can dominate raw yield trajectories, detrending is a standard step before interpreting weather-related yield variability [11], [12], [13].

Explainable machine learning offers complementary tools for interpreting fitted models. SHAP provides additive feature contributions [14]; LIME offers selected local surrogate explanations [15]; model-reliance and permutation ideas support grouped dependence checks [16]; and ALE curves summarize response shape while reducing unrealistic feature combinations that can affect simple partial-dependence displays [17]. Rather than treating any single method as decisive, this study compares whether several methods point to the same physically meaningful driver groups.

## III. Data

The unit of analysis is a crop–state–year. The processed frame contains 1,257 observations from 1990–2025 for Barley, Canola, Oats, and Wheat across 12 U.S. states: Colorado, Illinois, Iowa, Kansas, Minnesota, Montana, Nebraska, North Dakota, Oklahoma, South Dakota, Texas, and Washington. Yield is measured in t ha$^{-1}$ from USDA NASS [18]. Weather features are derived from NASA POWER daily data [19] and summarized over growing-season windows labelled spring or winter.

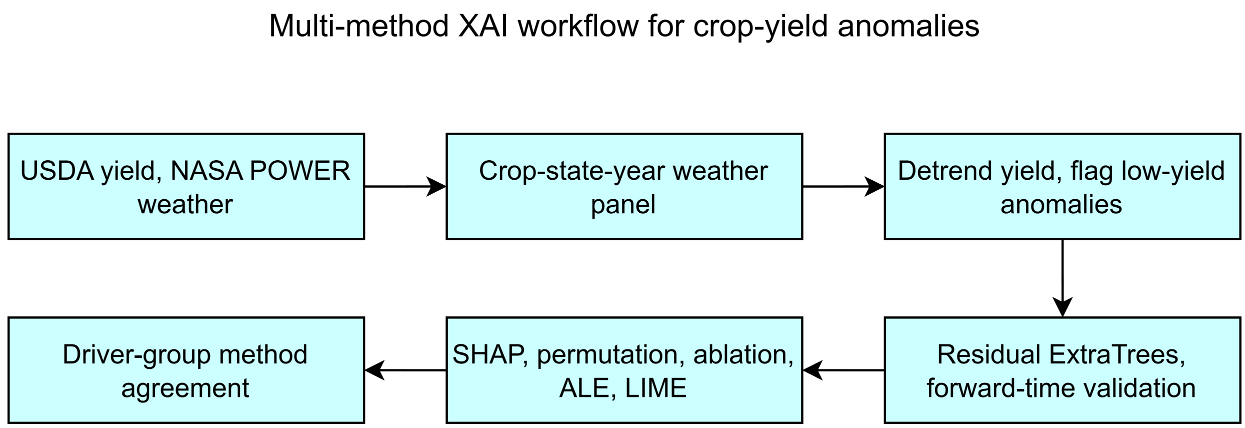


Fig. 1. Workflow for detecting detrended low-yield anomalies, fitting residual models, applying multiple explanation methods, and comparing driver-group support.

TABLE I
Dataset summary by crop.

| Crop | Obs. | States | Years | Anom. | Window |
|---|---|---|---|---|---|
| Barley | 283 | 10 | 1990–2025 | 52 | spring |
| Canola | 133 | 6 | 1991–2025 | 21 | spring |
| Oats | 416 | 12 | 1990–2025 | 71 | spring |
| Wheat | 425 | 12 | 1990–2025 | 70 | spring/winter |

Each row contains crop identity, state identity, year, growing-season window, yield, representative state latitude and longitude, and aggregated weather indicators. The weather extraction uses representative state coordinates rather than gridded or crop-area-weighted weather. Therefore, explanations should be interpreted at a state-representative scale; within-state heterogeneity in soils, irrigation, cultivar, and management is not resolved. The panel is moderately unbalanced: Canola has fewer observations and fewer states than Oats or Wheat, so crop-specific patterns should not be generalized equally across crops.

The XAI analysis uses 35 full-season weather features grouped into heat, drought, frost/cold, excess rain, and radiation. Numerical predictors are median-imputed and standardized inside the model pipeline; crop and state identifiers are imputed by most frequent category and one-hot encoded. The same preprocessing pipeline is used for yield and residual models. The feature set includes rainfall totals, wet days, dry spells, heat days, heatwave summaries, frost days, and radiation summaries.

## IV. Method

### A. Anomaly Screening

For each crop–state series, yield $y_{c,s,t}$ is linearly detrended against year:

$$y_{c,s,t} = \alpha_{c,s} + \beta_{c,s} t + e_{c,s,t}. \quad (1)$$

The residual is standardized within the crop–state series,

$$z_{c,s,t} = \frac{e_{c,s,t}}{\mathrm{sd}(e_{c,s,\cdot})}, \quad (2)$$

using sample standard deviation. An observation is flagged as a low-yield anomaly when $z < -1.0$. Robustness checks use stricter thresholds of $z < -1.5$ and $z < -2.0$, and compare the linear-trend anomaly set with a centered seven-year rolling-trend baseline.

TABLE II
Extreme-weather driver groups used in the grouped-XAI analysis.

| Group | Description | Features |
|---|---|---|
| Heat | High temperature, heatwave, and growing-degree exposure | 12 |
| Drought | Low rainfall, dry spells, and hot-dry compound stress | 8 |
| Frost/cold | Cold nights, frost days, and low minimum-temperature stress | 5 |
| Excess rain | Heavy rainfall, wetness, and short-duration rainfall maxima | 8 |
| Radiation | Seasonal solar-radiation anomalies | 2 |

TABLE III
Compact anomaly-screening robustness checks.

| Check | Setting | Result |
|---|---|---|
| Threshold | $z < -1.0$ | 214 anomalies; default broad anomaly set |
| Threshold | $z < -1.5$ | 85 anomalies; stricter low-yield subset |
| Threshold | $z < -2.0$ | 30 anomalies; most severe subset |
| Detrending | 7-year rolling | 192 anomalies; Jaccard overlap 0.618 |

### B. Yield and Residual Models

The yield baseline uses ExtraTrees with year, location, crop, state, and weather features [20], [21]. Its role is to check whether the data contain substantial predictive signal for yield levels. The residual model uses weather features, location, crop, and state to predict detrended yield residuals. Its role is narrower: it provides a controlled diagnostic surface for comparing explanation methods.

All SHAP-based explanations are computed on the fitted residual ExtraTrees model, not the yield-level baseline. Main global summaries use the forward-time evaluation split, while anomaly-focused summaries use low-yield anomalies within that split unless explicitly marked

TABLE IV
Forward-time yield and residual model performance. RMSE is in t $ha^{-1}$.

| Target | Protocol | Scope | $n$ | $R^2$ | RMSE |
|---|---|---|---|---|---|
| Yield | train 1990–2015; test 2016–2021 | all | 209 | 0.807 | 0.527 |
| Yield | train 1990–2018; test 2019–2025 | all | 236 | 0.813 | 0.529 |
| Residual | train 1990–2015; test 2016–2025 | all | 343 | 0.091 | 0.431 |
| Residual | train 1990–2015; test 2016–2025 | anom. | 67 | -4.396 | 0.660 |

as selected LIME cases. This separation prevents global feature importance from being interpreted as event-level anomaly explanation.

## C. Global Explanation Methods

Let $F_g$ be the feature set for driver group $g$, and let $\phi_{i,j}$ be the SHAP value for observation $i$ and feature $j$. Grouped SHAP magnitude is

$$G_{i,g}^{abs} = \sum_{j \in F_g} |\phi_{i,j}|. \tag{3}$$

Global grouped SHAP is the mean of $G_{i,g}^{abs}$ across forward-time evaluation rows; anomaly-only grouped SHAP averages the same quantity over anomaly rows in the same evaluation split. Signed grouped SHAP,

$$G_{i,g}^{signed} = \sum_{j \in F_g} \phi_{i,j}, \tag{4}$$

is retained for directional inspection, while the main rankings use absolute grouped SHAP magnitude.

Group permutation jointly shuffles all features in $F_g$ across rows in the evaluation set while leaving all other features unchanged. Importance is the increase in residual RMSE relative to the unshuffled model:

$$I_g^{perm} = \mathrm{RMSE}(\hat{f}(X_{perm(g)}), y) - \mathrm{RMSE}(\hat{f}(X), y). \tag{5}$$

Group ablation is a retraining-level diagnostic. For each driver group, all features in $F_g$ are removed, the residual model is retrained with the same split and hyperparameters, and importance is

$$I_g^{abl} = \mathrm{RMSE}(\hat{f}_{-g}(X_{-g}), y) - \mathrm{RMSE}(\hat{f}(X), y). \tag{6}$$

Permutation measures prediction-time dependence of the fitted model; ablation measures whether the model can compensate after a group is removed and retrained.

## D. Event-Level Explanations, ALE, and Agreement

Event-level evidence is restricted to two non-perturbation views. First, grouped SHAP is aggregated for each anomaly event and the largest grouped contribution is used as the event's top driver. Second, LIME is applied only to selected case-study events from 2012, 2021, and 2022; selected cases are the lowest-$z$ anomaly rows within those years. LIME is therefore a selected-case check rather than a full-dataset ranking method.

TABLE V
Explanation methods and their role in the diagnostic protocol.

| Method | Main question | Use in this study |
|---|---|---|
| SHAP | Which features contribute to a tree-model prediction? | Feature-level ranking and signed residual inspection. |
| Grouped SHAP | Which physical driver group has the largest total contribution? | Global and anomaly-row driver ranking. |
| Permutation | How much does error rise when a group is jointly shuffled? | Prediction-time dependence check. |
| Ablation | What is lost when a group is removed and the model is retrained? | Retraining-level model-reliance check. |
| ALE | How does model response change across exposure levels? | Response-shape plausibility check. |
| LIME | How is a selected event explained locally? | Selected 2012, 2021, and 2022 case checks. |

ALE curves are used as response-shape diagnostics rather than causal dose-response estimates. They are preferred over simple PDP-style displays because many weather features are correlated. The curves help check whether the fitted residual model responds plausibly to selected heat, drought, and rainfall variables, but they do not identify independent causal effects.

For a scope $s$ and method $m$, let $T_m(s)$ be the rank-one driver group. The consensus vote count for driver group $g$ is

$$V_g(s) = \sum_m \mathbf{1}\{T_m(s) = g\}. \tag{7}$$

We use rank-one vote counts rather than averaging raw explanation scores because SHAP magnitudes, RMSE-increase diagnostics, and LIME scores are measured on different scales. The vote count is therefore a scale-free summary of whether methods select the same top driver group.

# V. Results

## A. Anomalies and Model Skill

The default threshold identifies 214 low-yield anomalies, or 17.0% of the processed panel. The threshold is intentionally inclusive: it is designed to collect a broad set of below-trend yield events for explanation rather than only the most extreme failures. Tightening the threshold produces smaller nested sets, while the rolling trend baseline shows moderate overlap with the linear-trend anomaly set (Table III). We therefore interpret the reported XAI results as conditional on the explicit linear detrending choice, while using the stricter thresholds and rolling-trend comparison as robustness checks rather than alternative main definitions.

Figure 2 shows that the anomaly set is not evenly distributed over time. The highlighted years are retained as sanity-check years because broad heat and drought conditions were documented for 2012 and for parts of the Northern Plains and western U.S. in 2021–2022 [22], [23], [24]. Table IV shows why the residual model is interpreted

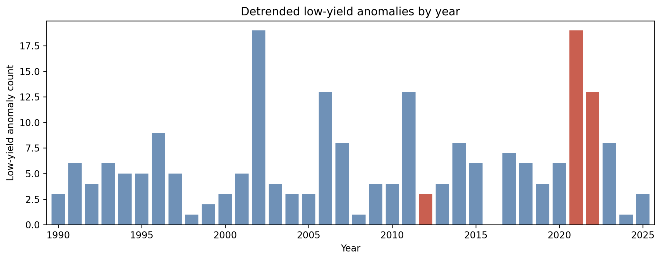


Fig. 2. Detrended low-yield anomaly counts by year, with 2012, 2021, and 2022 highlighted as plausibility-check years rather than training labels.

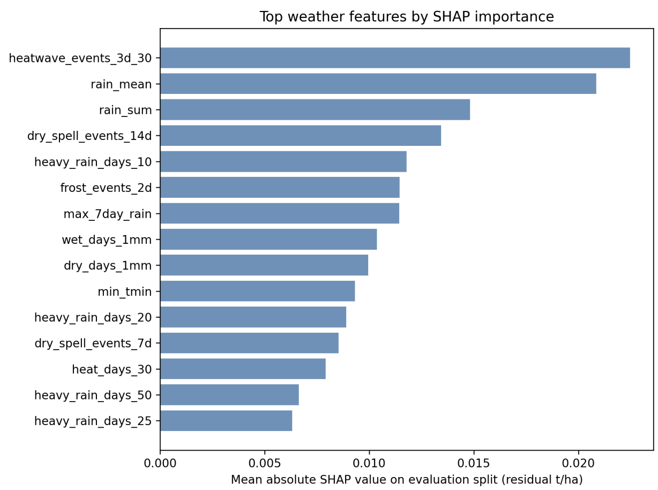


Fig. 3. Top weather features by mean absolute SHAP value in the residual model on the forward-time evaluation split.

cautiously: yield-level prediction is strong, but residual anomaly-magnitude prediction is weak, so the XAI results are diagnostic explanations of fitted-model behavior.

## B. Feature and Driver Rankings

Figure 3 indicates that the residual model relies on a mixture of rainfall, dry-spell, heatwave, wetness, and cold-stress features rather than a single variable family. This motivates grouping features by physical driver process instead of interpreting isolated feature rankings alone.

Grouped SHAP shows a shift between forward-time evaluation rows and anomaly rows (Fig. 4). On the forward-time anomaly subset, heat and drought are the two largest grouped contributions, with excess rain next. The difference between evaluation-row and anomaly-only summaries is important: anomaly explanation should not be reduced to global feature importance alone.

Table VI provides complementary global and anomaly-focused views. Grouped SHAP, permutation, and ablation are all reported on the forward-time evaluation split; LIME remains a selected-case check and is identified by its smaller evaluation scope. Because the score units differ across methods, the table should be read as a cross-method rank summary rather than an identical-scale statistical comparison. Globally, heat is the strongest group under

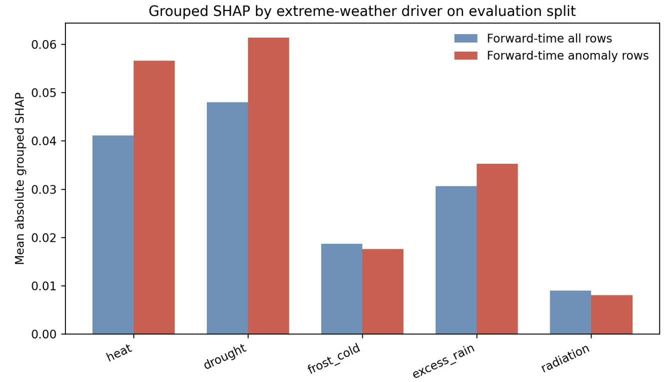


Fig. 4. Grouped SHAP summaries for forward-time evaluation rows and anomaly rows.

TABLE VI
Global and anomaly-focused driver-group rankings across explanation methods.

| Scope | Method | Rank 1 | Rank 2 |
|---|---|---|---|
| Global | Ablation | heat (0.0078) | excess rain (0.0025) |
| Global | Permutation | heat (0.0214) | drought (0.0199) |
| Global | Grouped SHAP | drought (0.0480) | heat (0.0411) |
| Anomaly | Ablation | heat (0.0287) | radiation (0.0044) |
| Anomaly | Permutation | drought (0.0243) | excess rain (0.0113) |
| Anomaly | Grouped SHAP | drought (0.0614) | heat (0.0566) |
| Anomaly | LIME cases | heat (0.0238) | drought (0.0224) |

retraining ablation and group permutation, while grouped SHAP ranks drought first. On anomaly rows, heat and drought become co-leading signals: heat ranks first under group ablation and selected LIME cases, while drought ranks first under grouped SHAP and group permutation.

Figure 5 explains why method comparison is useful. Permutation measures how much the fitted model depends on a driver group at prediction time; ablation measures whether the model can compensate when the group is removed and retrained. Heat ranks highest under both global diagnostics, suggesting robust global reliance. Drought ranks strongly under permutation but less strongly under retraining ablation, suggesting that drought information partly overlaps with other rainfall and wetness features.

## C. Response Curves and Agreement

ALE curves provide qualitative support for the driver rankings. Heat-related curves generally decline at higher exposure levels, consistent with adverse high-temperature behavior in the fitted residual model. Rainfall and dry-spell curves are more nonlinear, suggesting that the model does not treat water stress as a simple monotonic process.

Table VII summarizes rank-one support across heterogeneous explanation methods. It should be read as a scale-free consensus summary, not as a statistical test of driver importance. Figure 7 visualizes rank positions rather than method-by-method rate estimates: darker/high-rank cells indicate which driver groups are placed near the top by each method. Global diagnostics give majority rank-one

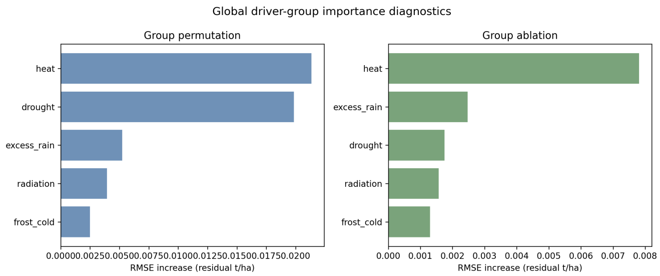


Fig. 5. Global group permutation and group ablation importance measured by residual RMSE increase.

TABLE VII
Scale-free rank-one support across heterogeneous explanation methods.

| Scope | Methods | Driver | Interpretation |
|---|---|---|---|
| Global | 3 | heat, 2/3 | Majority support from group ablation and group permutation. |
| Anomaly | 4 | heat, 2/4; drought, 2/4 | Co-leading anomaly-focused signals under rank-one support. |

support to heat, while anomaly-focused explanations split their rank-one support between heat and drought.

Table VIII reports representative cases from the selected LIME case-study set and illustrates selected case-level method sensitivity. Its purpose is not to validate event-level correctness. Agreement between grouped SHAP and LIME in four of six selected cases provides stronger diagnostic support for those examples, while the two disagreements indicate that event-level driver assignments should be treated cautiously and not reported as robust single-driver claims.

## VI. Discussion

The multi-method comparison supports a more nuanced conclusion than a single ranking would provide. Heat is the strongest global model-reliance driver: both group permutation and retraining ablation place heat first in the forward-time evaluation setting. Anomaly-focused explanations are more method-sensitive. Drought ranks first in evaluation-scope grouped SHAP and anomaly-row permutation, while heat ranks first in anomaly-row ablation and selected LIME cases. Excess rain is a meaningful secondary signal because rainfall and wetness features appear in feature-level and grouped summaries.

Method disagreement is informative. SHAP describes additive contributions within the fitted tree model; permutation measures prediction-time dependence; retraining ablation measures whether other variables can substitute for a removed group; ALE describes response shape; and LIME provides selected local surrogate checks. Weather features are correlated, so methods that answer different questions should not agree perfectly. The agreement protocol clarifies which driver claims are stable across several lenses and which are method-sensitive.

TABLE VIII
Selected event-level method-sensitivity checks using grouped SHAP and LIME.

| Crop/state | Year | $z$ | SHAP | LIME |
|---|---|---|---|---|
| Canola ND | 2012 | -1.49 | heat | heat |
| Oats KS | 2012 | -2.33 | heat | drought |
| Barley SD | 2021 | -2.98 | heat | heat |
| Wheat WA | 2021 | -2.92 | heat | heat |
| Oats OK | 2022 | -3.44 | drought | heat |
| Wheat NE | 2022 | -2.64 | drought | drought |

The practical application of the protocol is decision support rather than automatic event attribution. After a low-yield crop–state–year is detected, the workflow can help analysts inspect which driver groups the residual model emphasizes and whether that emphasis is stable across contribution, perturbation, retraining, curve-based, and selected local explanations. Such a tool can support post-season diagnosis, prioritize heat- or drought-related adaptation questions, and guide follow-up agronomic investigation. It should not be used as a direct insurance payout rule, farm-level recommendation engine, or proof that one weather driver caused a specific yield loss.

The global and anomaly-focused scopes should also be interpreted differently. Global model reliance summarizes behavior across all forward-time evaluation rows, including ordinary years. It therefore asks which driver group is generally important for the fitted residual model. Anomaly-focused explanation asks a more specific question: among the below-trend cases selected by the anomaly screen, which driver groups dominate the explanation summaries? The answer shifts from a clearer global heat signal to a heat–drought split on anomaly rows. This is a substantive result rather than a nuisance, because low-yield events can concentrate in regions of feature space where drought, rainfall, and heat variables overlap more strongly than they do in the full evaluation set.

The state-representative design is most appropriate for screening and triage. For example, if a region repeatedly shows below-trend yield years whose explanations are heat-dominant under both permutation and ablation, the result can motivate more localized investigation into heat-stress timing, cultivar sensitivity, and possible management adaptation. If grouped SHAP and LIME disagree on a selected event, the protocol deliberately lowers confidence in a single-driver narrative. This behavior is desirable for operational dashboards because it helps users separate robust model-reliance signals from event-level cases that require domain follow-up.

The analysis also provides a reproducible template for extending the study. Future versions can replace representative state coordinates with gridded or crop-area-weighted weather, add soil and irrigation covariates, or evaluate crop-specific calendars. The same agreement logic would still apply: claims should be strongest when

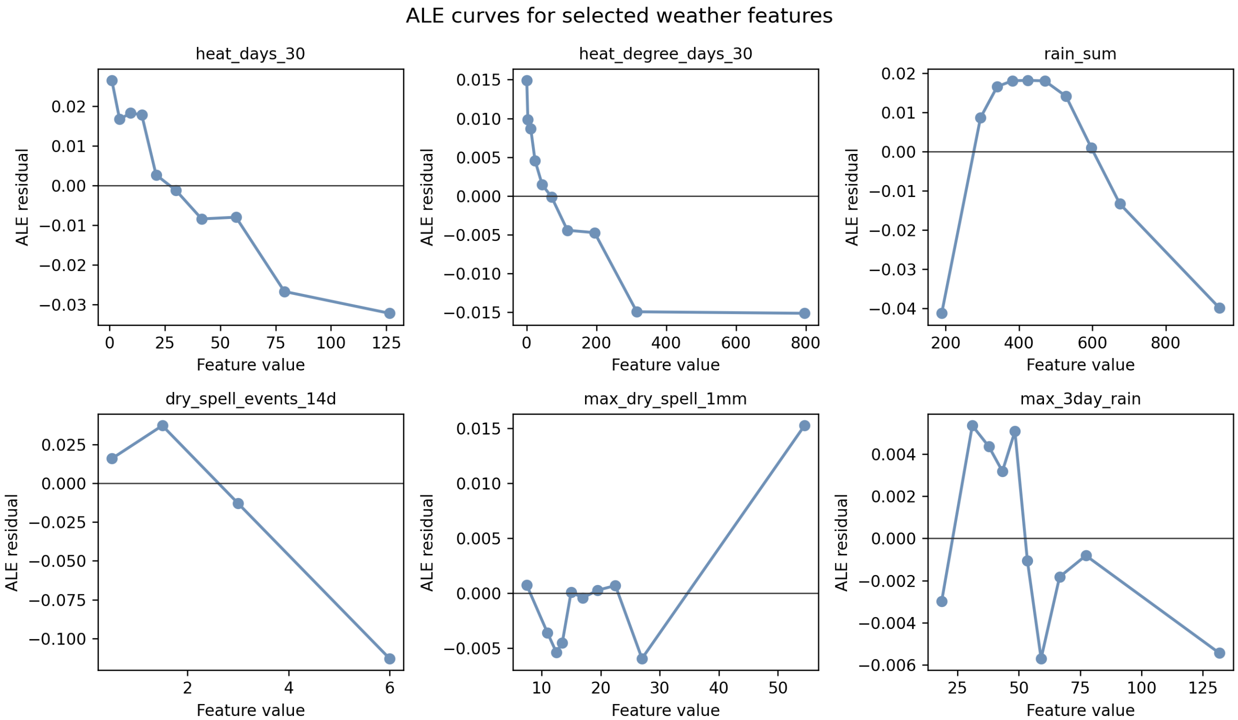


Fig. 6. ALE curves for selected heat, drought, and rainfall features.

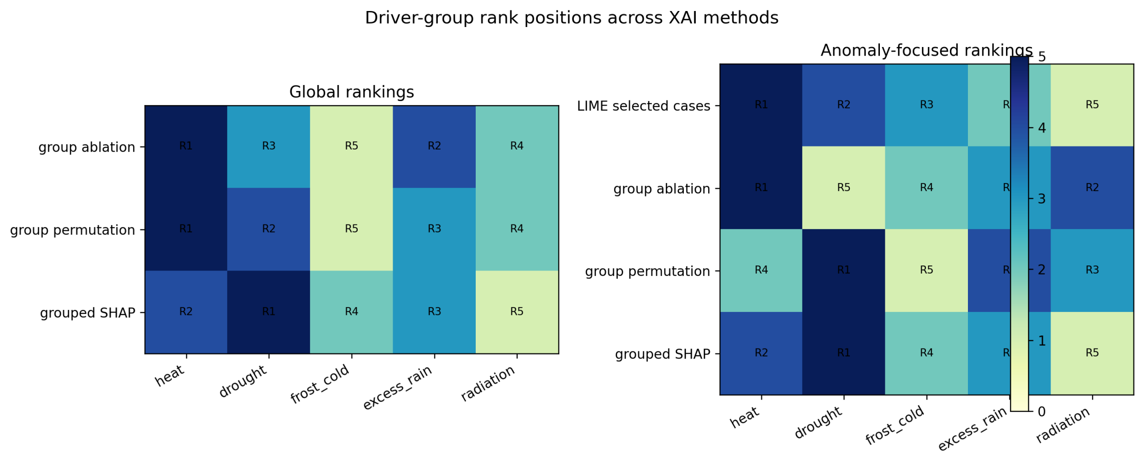


Fig. 7. Driver-group rank positions across global and anomaly-focused XAI methods.

feature contribution, perturbation, retraining, response-curve, and local explanation views point toward the same driver group. Conversely, method disagreement should be reported as uncertainty rather than hidden by averaging incompatible scores.

Several limitations remain. The dataset is state-level and uses representative coordinates, so it cannot capture within-state weather, soil, irrigation, cultivar, or management heterogeneity. Detrending choices influence which rows are labelled as anomalies. The residual model has limited skill for anomaly magnitudes, which narrows the claim to diagnostic explanation of fitted-model behavior. Finally, the analysis is not a formal climate-event attribution study; that task requires a different causal and probabilistic framework [25], [26], [27].

## VII. Conclusion

This study presents a reproducible multi-method XAI protocol for explaining detrended crop-yield anomalies. By combining anomaly screening, residual modeling, SHAP, grouped SHAP, group permutation, group ablation, ALE, selected LIME cases, and method agreement, the analysis avoids relying on a single explanation method. The main empirical pattern is that heat is the strongest global model-reliance driver, while heat and drought are co-leading anomaly-focused signals depending on the explanation method. The approach provides a cautious, auditable bridge between crop-yield prediction and diagnostic decision support.

## References


[1] D. Paudel et al., "Machine learning for large-scale crop yield forecasting," Agricultural Systems, vol. 187, p. 103016, 2021.

[2] M. Meroni, F. Waldner, L. Seguini, H. Kerdiles, and F. Rembold, "Yield forecasting with machine learning and small data: What gains for grains?" Agricultural and Forest Meteorology, vol. 308–309, p. 108555, 2021.

[3] S. Khaki and L. Wang, "Crop yield prediction using deep neural networks," Frontiers in Plant Science, 2019.

[4] G. Leng and J. W. Hall, "Predicting spatial and temporal variability in crop yields: an inter-comparison of machine learning, regression and process-based models," Environmental Research Letters, vol. 15, p. 044027, 2020.

[5] C. Lesk, P. Rowhani, and N. Ramankutty, "Influence of extreme weather disasters on global crop production," Nature, vol. 529, pp. 84–87, 2016.

[6] M. Zampieri, A. Ceglar, F. Dentener, and A. Toreti, "Wheat yield loss attributable to heat waves, drought and water excess," Environmental Research Letters, 2017.

[7] E. Vogel et al., "The effects of climate extremes on global agricultural yields," Environmental Research Letters, 2019.

[8] M. Heino et al., "Increased probability of hot and dry weather extremes during the growing season threatens global crop yields," Scientific Reports, 2023.

[9] F. Schierhorn, M. Hofmann, T. Gagalyuk, I. Ostapchuk, and D. Muller, "Machine learning reveals complex effects of climatic means and weather extremes on wheat yields during different plant developmental stages," Climatic Change, vol. 169, p. 39, 2021.

[10] H. Sjulgard, T. Keller, G. Garland, and T. Colombi, "Relationships between weather and yield anomalies vary with crop type and latitude in sweden," Agricultural Systems, vol. 211, p. 103757, 2023.

[11] D. K. Ray, J. S. Gerber, G. K. MacDonald, and P. C. West, "Climate variation explains a third of global crop yield variability," Nature Communications, vol. 6, p. 5989, 2015.

[12] J. Lu, G. J. Carbone, and P. Gao, "Detrending crop yield data for spatial visualization of drought impacts in the united states, 1895–2014," Agricultural and Forest Meteorology, vol. 237–238, pp. 196–208, 2017.

[13] H. Meng et al., "Performances of different yield-detrending methods in assessing drought and flooding impacts on crop yield," Agricultural Water Management, vol. 296, p. 108795, 2024.

[14] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems, 2017.

[15] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should i trust you? explaining the predictions of any classifier," in Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016.

[16] A. Fisher, C. Rudin, and F. Dominici, "All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously," Journal of Machine Learning Research, vol. 20, no. 177, pp. 1–81, 2019.

[17] D. W. Apley and J. Zhu, "Visualizing the effects of predictor variables in black box supervised learning models," Journal of the Royal Statistical Society: Series B, vol. 82, no. 4, pp. 1059–1086, 2020.

[18] United States Department of Agriculture, National Agricultural Statistics Service, "Quick stats database and api," 2025, accessed 2026-06-15. [Online]. Available: https://www.nass.usda.gov/Quick_Stats/

[19] NASA Prediction Of Worldwide Energy Resources Project, "Power data access viewer and api," 2025, accessed 2026-06-15. [Online]. Available: https://power.larc.nasa.gov/data-access-viewer/

[20] P. Geurts, D. Ernst, and L. Wehenkel, "Extremely randomized trees," Machine Learning, vol. 63, no. 1, pp. 3–42, 2006.

[21] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[22] NOAA National Centers for Environmental Information, "Monthly drought report: Annual 2012," 2013, accessed 2026-06-15. [Online]. Available: https://www.ncei.noaa.gov/access/monitoring/monthly-report/drought/201213

[23] NOAA National Integrated Drought Information System, "2020-2021 drought in the u.s. northern plains and canadian prairies," 2022, accessed 2026-06-15. [Online]. Available: https://www.drought.gov/documents/2020-2021-drought-us-northern-plains-and-canadian-prairies

[24] NOAA National Centers for Environmental Information, "Monthly drought report: Annual 2022," 2023, accessed 2026-06-15. [Online]. Available: https://www.ncei.noaa.gov/access/monitoring/monthly-report/drought/202213

[25] F. E. L. Otto, "Attribution of weather and climate events," Annual Review of Environment and Resources, vol. 42, pp. 627–646, 2017.

[26] G. J. van Oldenborgh et al., "Pathways and pitfalls in extreme event attribution," Climatic Change, vol. 166, p. 13, 2021.

[27] F. E. L. Otto, "Attribution of extreme events to climate change," Annual Review of Environment and Resources, vol. 48, pp. 813–828, 2023.